

COOLHACKERS · BAND FACTORY · POCKETFUL TRACK

Pocketful Delivery Line

One spec in. A verified wallet service out, where money is never created, destroyed or spent twice, built by four Band seats.

coordinator

Claude Sonnet

tester

GLM-5.3-Flash

developer

GLM-5.3-Flash

reviewer

Claude Sonnet

Claude Code seats · OpenCode + Featherless seats

THE TASK

A wallet where money is never created, lost or spent twice

1	JSON API Idempotent writes on 5 paths, exact split rounding, atomic settlements, export/import	79% checks shipped
2	Browser UI + holds Recovery from lost responses and stale reads; authorizations with partial captures	35% checks shipped
3	Statements + corrections Effective vs recorded time, historical overdraft checks, frozen snapshots	9% checks shipped
4	Refunds + batch corrections Whole-settlement corrections with strict error precedence	16% checks shipped

HOW IT IS JUDGED

50%

Factory
generic mandates, how far it got, reusable FACTORY.md

25%

App
presentation-ready UI and maintainable code

25%

Agent teamwork
shared work, real review, no human steering

Submissions close **Mon Oct 5, 23:59 PDT**

Never created, never destroyed, never spent twice



Never created or destroyed

Balances always sum to the seeded total

Split shares sum exactly, whole minor units

Corrections keep every historical view balanced



Never negative

No balance below zero, even transiently

Available = total - held, never below zero

No past balance negative after a correction



Never spent twice

Concurrent identical requests: one 201

A request is paid at most once; each capture once

Replays after an upgrade move nothing

HOW THE FACTORY PROVES IT, EVERY STAGE

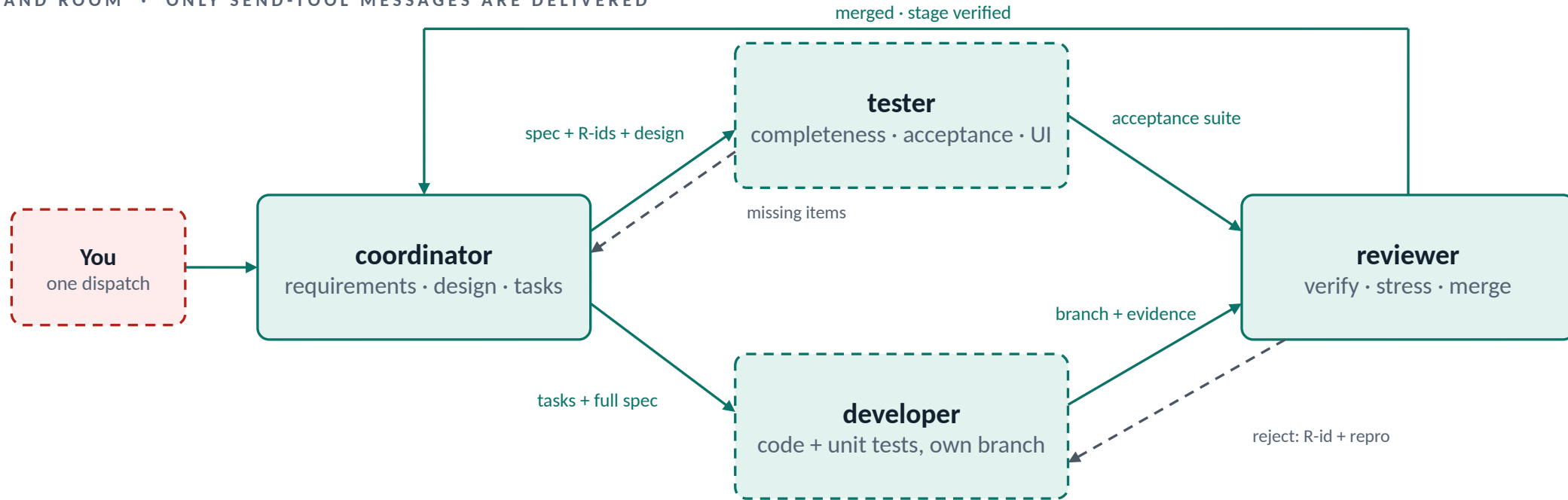
tester randomised concurrent and retried operations; invariants asserted after every run

reviewer independent stress run at stage verification, up to 50 requests in flight

design one critical section per write; exports carry idempotency records

Four seats in one Band room, one stage at a time

BAND ROOM • ONLY SEND-TOOL MESSAGES ARE DELIVERED



One acknowledgement
per handoff; no thank-you messages

Silent = working
wait 15 min, one identical resend

Never take over
another seat's work, ever

Latest handoff wins
stage + task id, supersedes earlier

 Claude Code seat

 OpenCode + Featherless seat

 Human: the dispatch only

Four seats, each owning one job and saying when it rejects

Seat	Harness	Model	Owns	Rejects when
coordinator	Claude Code	c laude - sonnet - 5	Requirements (R-ids), design, ADRs, tasks, reports. Never codes or merges	A handoff lacks spec, revision or evidence
tester	OpenCode	featherless/zai-org/GLM-5.3-Flash	Completeness review; acceptance suite per R-id; invariant attacks; UI at phone + desktop	A requirement has no test
developer	OpenCode	featherless/zai-org/GLM-5.3-Flash	Assigned tasks with unit tests, own worktree and branch	—
reviewer	Claude Code	c laude - sonnet - 5	Clean-copy review, only merger, isolated harness, invariant stress run	A check fails, an R-id lacks evidence, test-fitting

CUT FROM THE FIRST DESIGN, AND WHERE THE WORK WENT

architect → coordinator

developer-b → developer

qa-explorer → tester

release-sre-log-watcher → optional extension

Mandates are generated from one script and scanned with the harness's own vocabulary rules: 4 scanned, 0 problems.

Every rule in the spec becomes an R-id with a test



HOW BAD WORK GETS CAUGHT

Failure	Caught by
Missing requirement	tester's completeness review, before any code
Wrong behaviour	reviewer probes + tester's acceptance suite, independent of the developer
Integration breakage	reviewer's clean clone + isolated harness at stage verification
UI and recovery bugs	tester's browser checks at phone and desktop width
Broken money rule	tester's invariant attacks + reviewer's own stress run
Stuck work	the same R-id failing three times is split into smaller tasks
Test-fitting	reviewer rejects code that branches on test inputs

Three seats, one hour, \$12.40, and eight fixes

61 min

one dispatch, no steering

8/8

stage 1 checks, claimed stage 1

\$12.40

410 billed requests, GLM-5.2

186

turns; 75% routing and review

Finding	Fix now in every mandate
Seats read the supplied test files	Test folders and harness source unreadable in both permission files
Coordinator gave up after 2 minutes and wrote the code	Silent = working: wait 15 min, one identical resend, never take over
Replies sent as plain text never arrived	Only send-tool messages are delivered
Developer kept working on stale instructions	Every handoff names stage + task id and supersedes earlier ones
Coordinator merged and flip-flopped three times	Coordinator never codes or merges; the reviewer is the only merger
Thanks and acknowledgements woke seats up	One acknowledgement per handoff, nothing else

One container, designed for stage 4 from stage 1

Python 3.12 standard library

In-memory state, request queue of at least 256, one image with every asset bundled. No network at run time; 2 vCPU, 2 GiB, 50 requests in flight.

One critical section per write

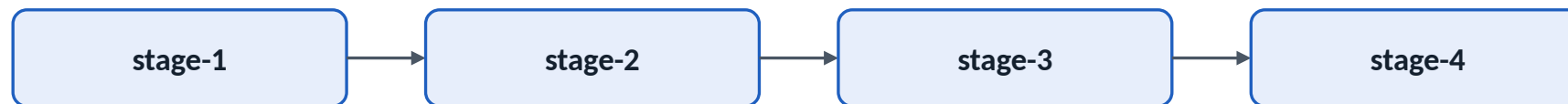
Parse → authenticate → resolve the idempotency key → validate → commit, under one lock. Password hashing runs outside it; export is an atomic snapshot.

Append-only payment ledger

Entries carry effective and recorded times, so one balance function answers "as of" and "as known at". Export is versioned; every stage imports every earlier one.

Stage-2 UI with an in-page script

Pay, refresh, request and split run from the page: a lost response shows uncertain and keeps the same key; the latest refresh wins; no reload needed.



Each folder is the previous one copied forward and extended. It passes every earlier suite and never the next one.

All of this lives in the dispatch message, never in a mandate: mandates stay generic.

Rehearsed once, redesigned, ready for a confirmation run

DONE

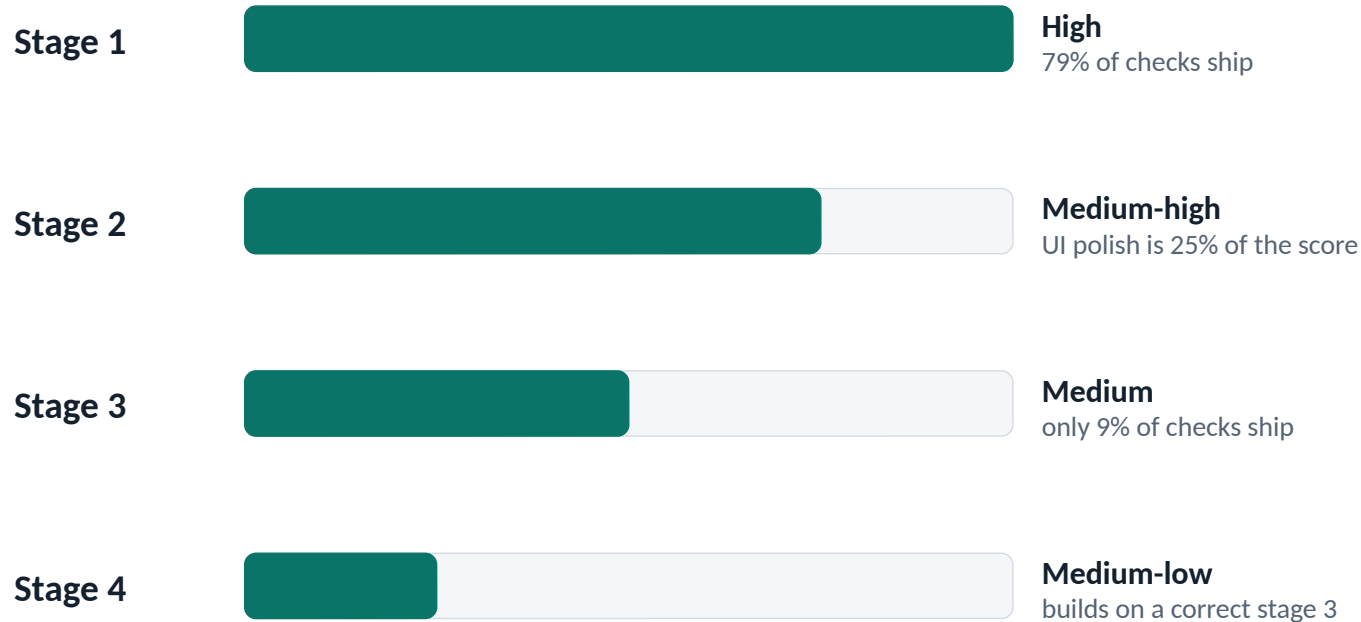
- Toy rehearsal: stage 1 verified, eight coordination findings
- Lean factory v2: four seats, mandates regenerated, scan 0 problems
- Test files and harness source blocked for both harnesses (tested)
- setup-seats.sh creates Claude Code and OpenCode seats; dry run passes
- Dispatch: money invariants proven by tester and reviewer
- Export carries idempotency records and tokens
- Lockfile moved to public PyPI

TO DO

- Team review of the lean-factory-v2 branch
- Fix the Claude login for the two Claude Code seats
- Create the four seats; delete the old six
- Confirmation run: toy stage 1 only, measure the cost
- Top up Featherless from that measurement
- Practice Pocketful stages 1-2, then freeze mandates
- README.md, FACTORY.md with measured cost and time, video

Judged run on Oct 2, with a day of buffer

CONFIDENCE BY STAGE (JUDGEMENT, NOT MEASURED)



The deciding risk is autonomy: one seat stalled on a permission prompt, or a rate limit on the shared subscription, spoils the judged run.

SCHEDULE



What the hackathon enforces, and what our setup must avoid

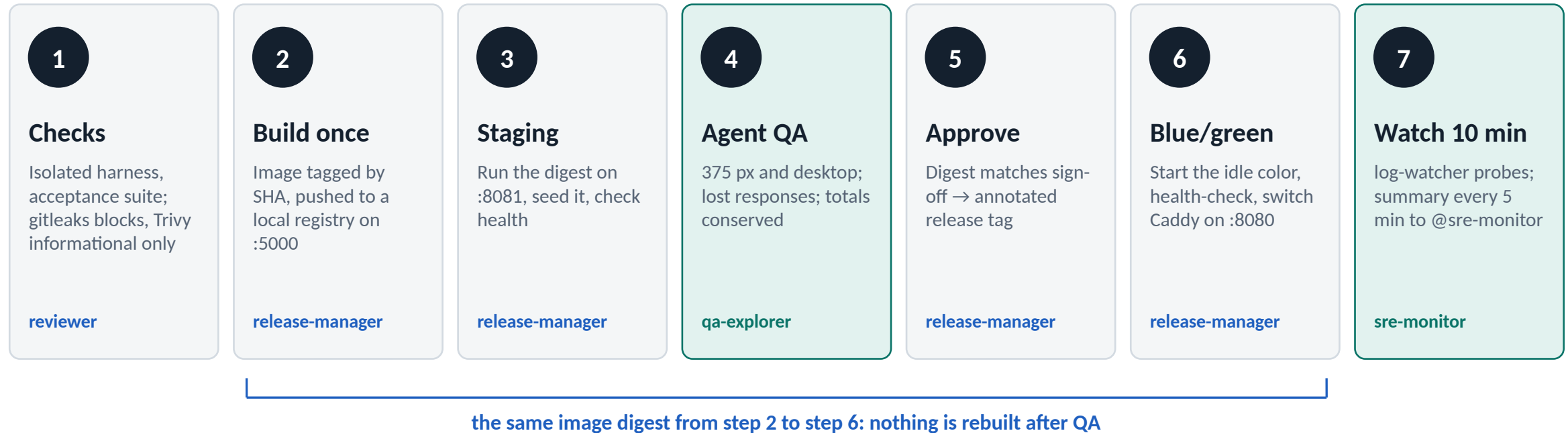
HACKATHON RULES · FROM THE PARTICIPANT GUIDE

Four gates 3+ seats with mandates · two-way @handle exchange · stage-1 builds · generic mandates	not ranked
Code written to the tests Build to the spec; shipped checks are only a signal	disqualified
Track words in a mandate No endpoints, fields or error codes; scan after every edit	disqualified
Hand-written code Every commit traces to a handoff in the room	stage doesn't count
Human steering One dispatch; no approvals, hints or reruns	lowers the score

OUR RISKS

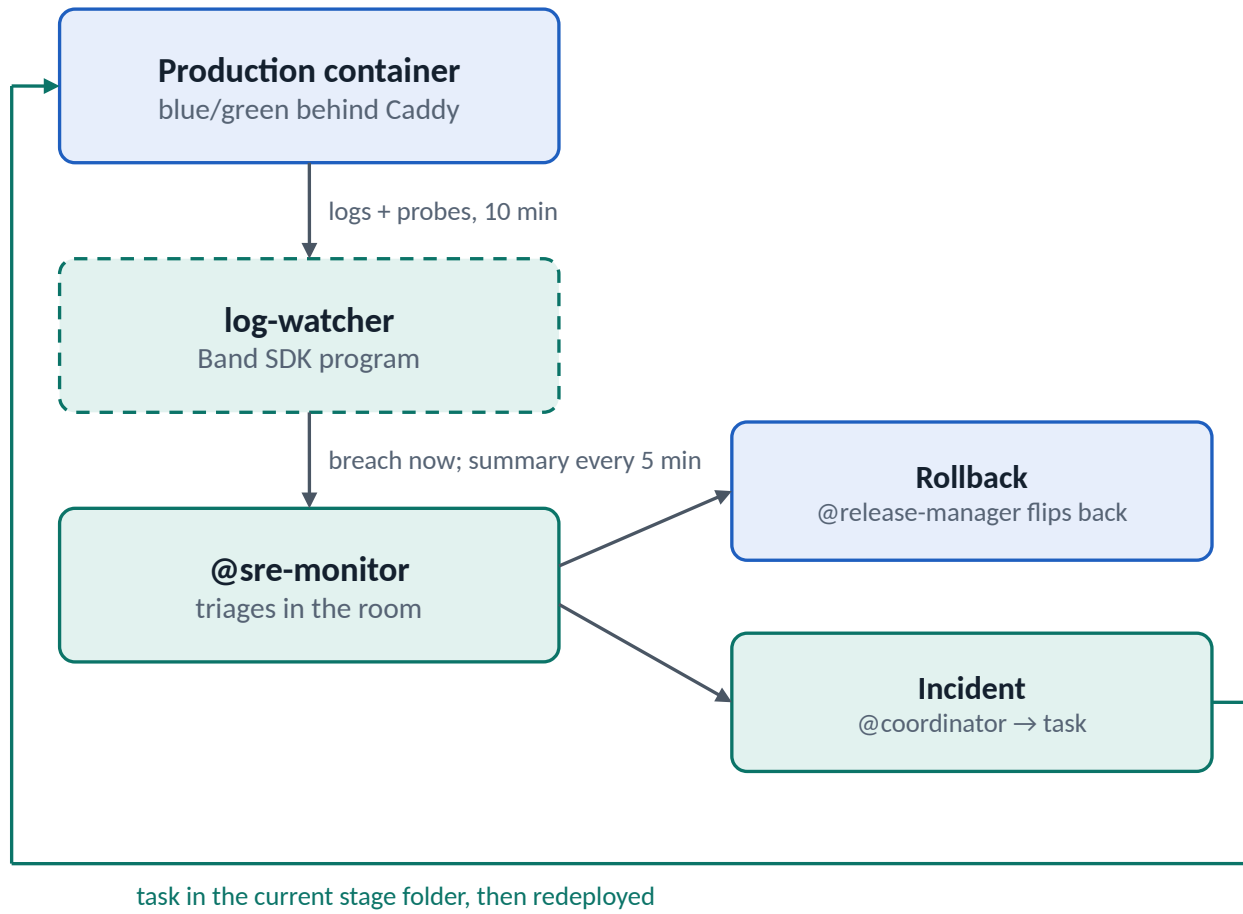
Docker Sandboxes All seats on host Docker, one shared repo
room.json is public No credentials in messages; seats can't push
Two Claude seats, one Pro plan Usage limits can stall a long run
Featherless budget Size the top-up from the confirmation run

Release pipeline, for when a project really deploys



Why it is not built. The specs require only a Dockerfile, RUN.md, a health check and no 5xx, and the rubric scores none of this. FACTORY.md documents it so another team can add these seats.

Production monitoring, for when a project really deploys



Watcher stops when the coordinator posts the final report.

PROBES AND THRESHOLDS

- **Conservation**
wallet balances $\neq$ seeded total: any difference
- **Replay mismatch**
an idempotent replay returns a different body: any
- **Any 5xx**
the spec forbids them, even under load: any
- **Refusal spike**
insufficient-funds refusals $> 3\times$ the previous 5 min
- **Latency**
p99 above 4 s (requests time out at 5 s)

NEXT

From spec to watched production, without a human in the loop

Review lean-factory-v2

four seats, eight fixes, money invariants

Confirmation run

toy stage 1 only; check fixes and cost

Judged run on Oct 2

fresh room, fresh repo, one dispatch

Thank you · team CoolHackers